Biochemical Guardrails — Q-Twin

Draft prepared from the 45-chip raw dataset (14/15/20/21 Mar 2026). This is a first pass built directly off the computed numbers — Eva should read through, correct anything that conflicts with lab notes, and own the final judgment calls (flagged below).

> *Errata (fixed since the first draft): Delta-f for the probe/target stages was originally computed as each stage's own start-to-end drift. That was wrong — cross-checking against the lab's own "All results_PCA3.xlsx" (Mean&SE sheet) showed the correct formula is a stage's ENDPOINT frequency minus the PRIOR stage's endpoint frequency (e.g. probe-endpoint minus chi-endpoint). ingest_raw_curves.py has been fixed and this now reproduces the -62.96 Hz / 10 µM reference almost exactly (see Section 2). This also changed which chips are SUCCESS vs FAILURE.*
>
> *Update: chip 15Mar_No.16 is now marked EXCLUDED, not FAILURE (~20,600 Hz nonsensical jump between its CHI and probe stages, a lab-flagged bad measurement, absent from the lab's own Success-rate tally too). 44 of 45 chips are valid: 30 SUCCESS, 14 FAILURE. The kinetic biomarker (Section 5) has also been redefined as early displacement, replacing the original slope version which was confirmed not predictive — see Section 5(c).*

1. Concentration bounds

Valid range: 0.5–10 µM. Concentrations outside this range are not physically meaningful for this assay and should be excluded from training data for the AI models in Weeks 2+.

DRAFT — needs Eva's review: this range is taken directly from the Week 1 guide (Task 5, item 13). The raw dataset also includes 20 µM, 40 µM and NC chips, used here only to characterize the crowding inversion (Section 2) and negative-control behavior (Section 4) — confirm this is the intended scope, not a contradiction.

2. The 10 µM crowding inversion

Above roughly 10 µM (probe concentration during the probe-deposition stage), Δf stops getting more negative and starts trending positive — surface crowding limits further mass loading, and non-specific/multilayer effects can push the frequency back up instead of down.

Reference number from the published paper/summary: −62.96 Hz at 10 µM, room temperature.

Concentration	Chips (n)	Mean probe Δf (Hz)	Range (Hz)	Note
0 µM	3	+10.40	+6.28 to +15.65	baseline, no probe
5 µM	3	-11.58	-26.05 to +7.01	
10 µM	3	-62.04	-105.86 to -31.61	peak binding
20 µM	3	-55.30	-120.81 to -10.36	above crowding onset
40 µM	3	-43.60	-76.77 to -10.07	

> *Table computed from the 14 Mar concentration-sweep chips (probe-endpoint minus chi-endpoint, averaged per chip; n=3 per concentration). The 10 µM mean, -62.04 Hz, now matches the -62.96 Hz reference almost exactly (the ~1 Hz residual is well within instrument noise / endpoint-selection rounding). The concentration trend is also physically clean: strongly positive at 0 µM (no probe, no mass added), peak binding at 10 µM, then decreasing magnitude at 20-40 µM — consistent with the crowding-inversion story, though note it never goes positive again at 20-40 µM in this dataset (magnitude just shrinks), so 'inversion' may be too strong a word for what's really 'crowding-limited plateau'.*

This resolves clarifying_questions.md item 2 — the mismatch was a bug in the original Δf computation (see Errata at top), not a data or lab-notes issue. Fixed in ingest_raw_curves.py.

3. Chip failure logic

SUCCESS if probe-stage Δf < 0 Hz. FAILURE otherwise (Δf ≥ 0, or missing probe data). Δf here is probe-endpoint minus chi-endpoint (see Errata).

Across the 44 valid chips (15Mar_No.16 excluded, see note above): 30 SUCCESS, 14 FAILURE (68.2% success rate). Mean probe Δf for SUCCESS chips is -75.94 Hz (range -372.07 to -0.18); mean for FAILURE chips is +21.36 Hz (range +0.10 to +104.30). The two groups are well separated — the closest chips to the 0 Hz boundary are 21Mar_No.9 (-0.18 Hz, SUCCESS) and 21Mar_No.8 (+0.10 Hz, FAILURE), about 0.3 Hz apart, still supporting 0 Hz as a clean cutoff.

DRAFT — needs Eva's review: confirm 0 Hz (not some small negative buffer, e.g. -0.5 Hz, to account for instrument noise) is the right cutoff — 21Mar_No.9 and 21Mar_No.8 sit close enough to zero (0.3 Hz apart) that measurement noise could flip either call. Also note the 68.2% chip-level rate is close to, but not identical to, the teacher's 70.18% (40/57) — see clarifying_questions.md item 1, likely a chip-count vs trial-count difference, not a real disagreement (Success rate sheet in All results_PCA3.xlsx counts 57 individual Probe-Chi/Target-Probe trial values, not 45 physical chips).

4. Real negative-control behavior

Chips No.7 and No.29 (21 Mar) are the only genuine 'no target present' runs. Plots: figures/NC_21Mar_No_7.png and figures/NC_21Mar_No_29.png.

With the corrected Δf, both NC chips are clearly FAILURE and clearly positive: No.7 is +3.29 Hz, No.29 is +22.615 Hz (previously this looked like a near-zero borderline call at +0.08 Hz under the buggy formula — the fix makes the negative-control result much cleaner and more convincing, not less). Draft description of the curves themselves: both NC chips show flat, low-amplitude, monotonic upward drift over their full run (roughly +10-15 Hz over 150-220s) with no sharp steps or large excursions — this looks like slow thermal/baseline drift, not a binding event. By contrast, a genuine hybridization curve (e.g. chip No.2, a SUCCESS chip from the same 21 Mar session) shows a fast initial drop of tens of Hz within the first ~20-30s and sharp step-like jumps later in the run, both absent from the NC curves. Chip No.29's probe-stage replicates are noisier than No.7's, with small dips instead of a smooth drift, but neither shows the large negative excursion characteristic of a SUCCESS chip.

DRAFT — needs Eva's review: this paragraph is Evin's read of the plots, standing in for Eva's Task 4 write-up. Eva should look at the actual figures (not just this description) and confirm/adjust — in particular the 'this is thermal drift, not binding' interpretation is an inference, not a lab-verified fact.

5. The Novel Kinetic Biomarker — REDEFINED as displacement, validated

(a) Time window

30 seconds, measured from the start of the probe-stage curve (also checked at 15s/45s/60s, see (c)).

(b) Definition — changed from slope to displacement

The original definition was the SLOPE of a linear fit to the probe curve's first 30s (binding_rate_probe_dfdt_30s in chip_summary.csv). That version is confirmed NOT predictive of outcome (see prior draft in version history / clarifying_questions.md item 8) — it's a same-stage-only quantity, mechanically disconnected from delta_f_probe, which is a cross-stage quantity (see Errata).

Redefined as early DISPLACEMENT: the probe curve's own value AT t=30s (linearly interpolated) minus the CHI-stage endpoint baseline — the same cross-stage baseline delta_f_probe itself uses, just read 30s into the run instead of at the end. New column: early_displacement_30s in chip_summary.csv.

(c) Validation

Correlation between early_displacement_30s and the corrected delta_f_probe is 0.9997 across the 44 valid chips — this is essentially an early readout of the same signal, which is exactly why it works as a feature (unlike the slope version, which was a genuinely different quantity). Simple threshold classification (predict SUCCESS if displacement < 0) scores 97.5% (39/40 scoreable chips) at the 30s window in this implementation, and holds flat at 97.5% across 15s/30s/45s/60s windows too.

> *These numbers were independently re-derived, not copied from Eva's message — the exact figures (97.5% here vs Eva's reported 97.7%/15s, 95.5%/30s, 93-98% range) differ slightly, most likely from a difference in interpolation or accuracy-scoring method between the two implementations. Both independently land on the same conclusion: displacement is a strong, usable feature; slope was not. Worth a quick sync between the two methodologies before Week 2, but not blocking.*

The one thing to watch: chip 21Mar_No.29 (true negative control) still shows an early value consistent with weak binding under the old slope metric, but the corrected full-run Δf (+22.615 Hz) is unambiguous FAILURE, and the new displacement metric agrees with the outcome for this chip. The previously-flagged single-replicate noise issue was specific to the slope calculation, not the displacement one.

Sources

- Computed values: qtwin/data/chip_summary.csv (45 chips, all 4 sessions).
- Cross-check against Eva's chip_index.csv: qtwin/data/cross_check_report.csv (0 mismatches).
- Plots: qtwin/figures/ (NC_21Mar_No_7.png, NC_21Mar_No_29.png, and grouped condition plots).